

# Ayna Musayeva

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## PROFILE

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Investigative writer specializing in undisclosed financing, ownership opacity, and the regulatory loopholes that enable illicit financial activity in the global art market, delivering original, previously unreported findings through a comprehensive investigative series. AI engineer architecting full-stack, retrieval-augmented research infrastructure — including Nimillux, a production-grade AI platform with built-in hallucination detection and citation tracking across large-scale document corpora.

## SELECTED RESEARCH & TECHNICAL PROJECTS

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### Capital, Ownership, and Opacity — Investigative & Regulatory Research Initiative (2024 – Present)

*A comprehensive investigative and regulatory research initiative examining ownership, financing, and systemic opacity across the global art market*

- Spearheaded a landmark, previously unreported investigative account of undisclosed family ties connecting senior auction-house executives to political leadership
- Performed clause-level statutory and regulatory analysis across international, federal, and art-market-specific legal frameworks, identifying and annotating the precise legislative wording that leaves illicit financial activity unaddressed
- Led a comprehensive legal and historical analysis of international art-restitution law, revealing a systemic loophole — recurring across every major treaty and legislative instrument — that lets ownership vanish into opacity through an ordinary sale rather than theft
- Conducted a rigorous, independent audit of widely cited market statistics, excluding figures with no verifiable source or methodology
- Identified and disclosed a material, unresolved sourcing conflict within a central historical account
- Conducted a formal regulatory-classification analysis of art investment funds spanning multiple legal frameworks — U.S. federal securities law under the SEC v. Howey Co. test, art-fund-specific regulatory regimes, and UNESCO cultural-property conventions
- Delivered a comprehensive, cross-jurisdictional analysis of fund structuring, registration, and tax implications across major offshore jurisdictions
- Exposed a systemic disclosure and oversight deficiency, rooted in permissively worded regulation, that leaves art investment funds outside the oversight applied to hedge funds despite structurally comparable architecture

### Nimillux — AI Platform (2024 – Present)

*A production-grade, fully local AI research platform with built-in hallucination detection and citation tracking across large-scale document corpora*

- Independently sourced, vetted, and curated a research corpus of tens of thousands of documents spanning a sprawling range of technical and financial disciplines, applying the same primary-source research discipline as the investigative work above — conducted entirely without external or AI assistance
- Directed the architecture of a fully local AI infrastructure with zero external API dependency, eliminating third-party data exposure and provider-side content-filtering risk
- Architected a hybrid retrieval layer combining dense semantic search with lexical matching and domain-aware routing, engineered to prevent any single subject area from dominating results across a large, heterogeneous multi-domain corpus

- Engineered a dual-layer hallucination-detection system, independently verifying each generated claim against its source and building a second, architecturally distinct verification model to close shared blind spots between retrieval and verification
- Identified a critical blind spot in the system's binary pass/fail verification model and drove the redesign into a tiered confidence framework, after an audit revealed it was masking weakly-supported claims inside answers reported as fully verified
- Directed resolution of a systemic class of query-understanding failures — spanning negation, imperative phrasing, symbolic notation, and typographical variance — that were causing valid queries to silently fail
- Built and maintained a hand-labeled regression-testing suite that grew continuously alongside the platform, catching and rejecting a flawed pattern-matching implementation before it reached production
- Designed a pre-generation trust-gating layer that abstains from responding entirely when retrieval confidence is insufficient, rather than allow the model to hallucinate around irrelevant context
- Built a dedicated citation-validation layer that catches fabricated reference identifiers invisible to topic-level semantic grounding
- Established the platform's hardware operating envelope through direct empirical stress-testing of model capacity on consumer-grade infrastructure, rejecting assumed limits in favor of measured failure thresholds
- Defined requirements for a full-featured conversational interface with persistent multi-turn memory, cross-document synthesis, and a dedicated contradiction-detection mode

## PROFESSIONAL EXPERIENCE

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### Copywriter, Apogee Media (2023 – 2024)

- Produced content across publications, video scripts, and web platforms, partnering with marketing and design teams to maintain consistent brand voice across campaigns
- Adapted messaging and tone across audience segments and content formats, revising copy against stakeholder feedback and brand style guidelines under campaign deadlines

### Editor, Musée Magazine (2018 – 2019)

- Curated and edited the emerging-artist roster for the magazine's 21st issue, interviewing photojournalists and artists for weekly publication
- Managed freelance contributor relationships and the editorial production schedule
- Edited and fact-checked contributor submissions for publication, coordinating with design and photo teams to finalize layout under weekly deadlines

### Gallery Assistant, Frieze Art Fair (2019)

- Handled artwork installation, client-facing sales, and logistics for exhibiting galleries
- Maintained collector and artist contact records to support post-fair follow-up
- Coordinated with exhibiting galleries on artwork presentation, condition reporting, and insurance documentation to support fair-day logistics

## EDUCATION

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### MFA, New York Academy of Art (2018)

### BA, Western Caspian University (2016)

## CERTIFICATES

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**Documentary Practices and Visual Journalism — International Center of Photography (2022)**

**Creative Practices — International Center of Photography (2020)**

## LANGUAGES

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English (professional working proficiency) • Russian (fluent) • Turkish (fluent) • Azerbaijani (fluent)

## ADDITIONAL SKILLS

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**Research & Editorial:** primary-source verification, financial and regulatory analysis, bookkeeping and financial-statement review, academic citation, publication management

**Technical:** Python, retrieval-augmented generation (RAG) architecture, vector search (FAISS), OCR and document processing, local/offline LLM deployment